An experimental comparison of at-issueness diagnostics

Word count (including references, excluding abstract and supplements): 11,098

Abstract At-issueness is a key concept in semantics and pragmatics, distinguishing the main point of an utterance from background information. However, there is no consensus on how at-issueness should be defined, and different diagnostics for at-issueness reflect different assumptions about its underlying nature. This paper presents a systematic experimental comparison of five diagnostics of at-issueness, applied to the same target contents, including the contents of non-restrictive relative clauses and the complements of clause-embedding predicates. The results reveal substantial differences between diagnostics, indicating that they are not interchangeable. In particular, diagnostics differ in whether they detect at-issueness differences between particular contents, and diagnostics measuring the at-issueness of contents embedded in questions yield greater differentiation than diagnostics embedding them in assertions. This latter result suggests that the speech act used to present a content plays a key role in affecting diagnostic outcomes. We argue that this effect of speech act on at-issueness ratings is due to how questions and assertions relate at-issue and not-at-issue content to speaker commitments, and cannot be explained by different notions of at-issueness assumed in the literature.

Keywords: at-issueness, diagnostics, experimental pragmatics, non-restrictive relative clauses, clause-embedding predicates

1 Introduction

At-issueness is a key concept in theoretical semantics and pragmatics,[1] used in the analysis of various phenomena including presupposition, conventional implicature, evidentials, expressives, NPI licensing, and co-speech gestures (e.g., Faller 2002; 2019; Horn 2002; Potts 2005; Simons et al. 2010; Lee 2011; Tonhauser et al. 2018; Esipova 2019; Korotkova 2020; Barnes & Ebert 2023). It is generally understood as distinguishing the main point of an utterance (at-issue content) from propositions conveying background information (not-at-issue content). However, there is no consensus on how at-issueness should be defined, and different diagnostics reflect different assumptions about its underlying nature (e.g., Tonhauser 2012; Snider 2017a; b; 2018; Tonhauser et al. 2018; Koev 2018; Faller 2019; Esipova 2019; Korotkova 2020). Consequently, empirical claims about the at-issueness of contents may be relative to specific diagnostics, and different diagnostics may not target the same underlying phenomenon.

1.1 Definitions of at-issueness

One influential definition assumes that discourse is structured around *Questions Under Discussion* (QUDs; Roberts 2012; Ginzburg 1996) and defines at-issue content as utterance content intended to address a QUD (Amaral et al. 2007; Simons et al. 2010):

(1) QUD at-issueness: (based on Simons et al. 2010, p. 323)
A content m is at-issue in a context c iff
a. m is relevant[2] to the QUD in c, and

[1] Potts (2015) credits the term to Bill Ladusaw, "who began using it informally in his UCSC undergraduate classes in 1985" (p. 2).

b. the speaker has a recognizable intention to address the QUD via *m*.

Another prominent definition, given in (2), is grounded in understanding an assertion as a proposal to update the common ground (Stalnaker 1978; Clark & Schaefer 1989; Ginzburg 1995; Farkas & Bruce 2010). Here, the at-issue content of an assertion is defined as the content that is proposed for addition to the common ground (Murray 2010; 2014; AnderBois et al. 2010; 2015, Koev 2013), while not-at-issue content is presupposed (i.e., already entailed in the common ground; Stalnaker 1973; 2002), or added to the common ground without being proposed (as in Murray 2010; 2014; AnderBois et al. 2010; 2015).

(2) Proposal at-issueness: (Koev 2013, pp. 50–51)
A proposition *p* is at-issue in a context *c* iff

a. *p* is a proposal in *c* and
b. *p* has not been accepted or rejected in *c*.

These definitions differ in their theoretical grounding and there is no consensus on whether they characterize distinct underlying notions of at-issueness or perspectives on a single phenomenon.[3] Some works suggest a unified view: if proposing content for the common ground is intrinsically linked to managing the QUD (e.g., Farkas & Bruce 2010), the proposal-based definition can be seen as a special case of the QUD-based definition, applied to assertions (AnderBois et al. 2015; Faller 2019). Others have argued that the definitions may correspond to distinct underlying phenomena, motivated by empirical differences between at-issueness diagnostics (Koev 2018).

1.2 At-issueness diagnostics

Diagnostics of at-issueness operationalize the theoretical notions in (1) and (2) via linking assumptions about how at-issue content behaves in discourse. These can be organized around three properties of at-issue content identified in Tonhauser (2012) as motivating diagnostic strategies:

(3) Properties of at-issue content (Tonhauser 2012)

a. at-issue content addresses the question under discussion (see also Amaral et al. 2007; Simons et al. 2010),
b. the at-issue content of questions determines the relevant set of alternatives, and
c. at-issue content can be directly assented or dissented with (see also Shanon 1976; Groenendijk & Stokhof 1984; Faller 2002; 2006; Murray 2010).

Four commonly used diagnostics that build on the properties in (3) are illustrated in (4–7) for sentence-medial non-restrictive relative clauses (NRRCs), which are typically taken to contribute not-at-issue content, following Potts 2005.

(4) QUD diagnostic

Nora: *What did Greg buy?*
Leo: *Greg, who bought a new car, is envied by his neighbor.*
Question to participants: How well does Leo's response fit Nora's question?

[2] *m* may be a proposition or a question denotation, and relevance to the QUD can be defined in terms of contextual entailment of a partial or complete answer (Roberts 2012; Simons et al. 2010), or based on inferences about the probability of QUD-answers (Büring 2003; Beaver & Clark 2008).

[3] Further definitions proposed in the literature include those based on restrictive vs. non-restrictive modification (Esipova 2019; 2021), and coherence-based discourse structure (Hunter & Asher 2016; Jasinskaja 2016; Koev 2018). Clarifying the relationships among the various proposed definitions of at-issueness remains an important question for future research.

(5) ‘asking whether’ diagnostic

Nora: *Is Greg, who bought a new car, envied by his neighbor?*

Question to participants: Is Nora asking whether Greg bought a new car?

(6) Direct-dissent diagnostic

Nora: *Greg, who bought a new car, is envied by his neighbor.*
Leo: *No, that’s not true, he didn’t buy a new car.*

Question to participants: How natural is Leo’s rejection of Nora’s utterance?

(7) ‘yes, but’ diagnostic

Nora: *Greg, who bought a new car, is envied by his neighbor.*
Leo: *Yes, but he didn’t buy a new car. /*
Yes, and he didn’t buy a new car. /
No, he didn’t buy a new car.

Task for participants: Choose the response that sounds best.

The QUD-diagnostic (4) (Amaral et al. 2007; Lee 2011; Tonhauser 2012; Chen 2024) targets property (3a) by testing whether the target content can felicitously address an established QUD. It presents the content in a response to a question that makes the the target content relevant and elicits ratings of question-answer fit. Responses are expected to indicate a good fit when the content can be construed as at-issue and a bad fit otherwise. Accordingly, responses are expected to indicate a bad fit for the appositive NRRC in (4).

The ‘asking whether’ diagnostic (5) (Tonhauser et al. 2018; Solstad & Bott 2024; Degen & Tonhauser 2025) targets property (3b) by probing whether the target content determines the alternatives of a polar question. It presents the target content in a polar question, and elicits judgments about whether the speaker is asking whether that content is true. These judgments are expected to be indicate that the speaker is asking about the target content for at-issue content and that the speaker is not asking about the target content otherwise. Accordingly, participants are expected to judge that the speaker is not asking about the NRRC content in (5).

The direct-dissent diagnostic (6) (Faller 2002; 2006; Papafragou 2006; Amaral et al. 2007; Murray 2010; 2014; AnderBois et al. 2010; 2015; Tonhauser 2012; Syrett & Koev 2015), and the ‘yes, but’ diagnostic (7) (Xue & Onea 2011; Cummins et al. 2013; Destruel et al. 2015) target property (3c). Both present the target content in a declarative sentence and assume that only at-issue content can be directly assented or dissented with, whereas dissenting with not-at-issue content requires more indirect discourse moves. The direct-dissent diagnostic elicits naturalness ratings for a response directly dissenting with the target content; the response is expected to indicate naturalness for at-issue content and unnaturalness for not-at-issue content. Accordingly, responses indicating unnaturalness are expected for the NRRC content in (6). The ‘yes, but’ diagnostic presents a forced choice between *‘no’*, *‘yes, but’*, and *‘yes, and’* when denying the target content. Under the assumption that the *‘yes’*-responses provide indirect denials suitable for dissenting with not-at-issue content (Xue & Onea 2011), participants are expected to prefer one of the *‘yes’*-responses when denying not-at-issue content, such as the NRRC content in (7). On the other hand, participants are expected to prefer the direct denial with *‘no’* when dissenting with at-issue content.[4]

[4] Although at-issue content permits direct dissent, it does not require it, and indirect dissent is also possible (see, e.g., Cummins et al. 2013 on Shanon’s 1976 “Hey, wait a minute” diagnostic). Work in conversation analysis suggests that preferences for direct versus indirect dissent vary with context and content. Direct dissent is often overall dispreferred (Sacks 1987; Brown & Levinson 1978; Pomerantz 1984; Locher 2004), but the strength of this preference can vary with cultural norms (e.g., Kakava 2002) or speaker gender (e.g., Tannen 1990). In contrast, direct dissent can be preferred in adversarial legal contexts (Atkinson & Drew 1979), in disputes (Kotthoff 1993; Kangasharju 2009), or in responses to self-deprecating statements (Pomerantz 1984).

1.3 Diagnostic differences and notions of at-issueness

The question whether different definitions of at-issueness characterize a single phenomenon or distinct ones is also reflected in competing views about whether diagnostics probe distinct underlying notions or provide different ways of probing a single, unified phenomenon.

Tonhauser (2012) suggests that diagnostics based on the three properties in (3) can be taken to probe a common underlying notion of at-issueness, based on the QUD-based definition.

In contrast, views that take different diagnostics to probe distinct phenomena are motivated by empirical divergences between them. For example, although appositive NRRCs are widely taken to contribute not-at-issue content, a forced-choice continuation study by Syrett & Koev (2015) found that sentence-final appositive NRRCs pattern as more at-issue than medial ones, under a version of the direct-dissent diagnostic, which targets property (3c). Snider (2017b) argues that diagnostics targeting property (3c) are sensitive to this medial vs. final distinction, but diagnostics targeting properties (3a) or (3b) are not. Similar diagnostic differences have been reported for sentence-final slifting parentheticals (Koev 2018) and evidential inferences (Korotkova 2020).

Such observations have been interpreted to indicate that the different diagnostics do not diagnose the same underlying concept. For Koev (2018), the QUD-based notion of at-issueness and the proposal-based notion of at-issueness are distinct notions, and the corresponding diagnostics therefore diagnose distinct discourse phenomena. Other works (Snider 2017a; b; 2018; Korotkova 2020) maintain that diagnostics targeting properties (3a) and (3b) genuinely probe at-issueness, whereas diagnostics targeting property (3c) are instead sensitive to the availability of the target content for propositional anaphora. These latter diagnostics are taken to reflect whether the target content can provide anaphoric antecedents for response particles (*yes/no*) or propositional pro-forms (e.g., *that* in *that's not true*), which are present in the direct-dissent and 'yes but' diagnostics.

Although empirical differences between at-issueness diagnostics have been reported across a range of constructions, they have not, to date, been evaluated in a systematic experimental comparison that applies multiple diagnostics to the same contents. Moreover, there is currently no agreed-upon definition of at-issueness, and existing diagnostics may or may not be probing the same underlying phenomenon. This raises two central questions:

(8) Research questions

a. Do different diagnostics yield consistent results when applied to the same contents?
b. Do different diagnostics reflect distinct underlying notions of at-issueness?

1.4 Previous findings

Initial experimental work on the research questions in (8) has yielded mixed answers. As noted above, Syrett & Koev (2015) found a positional effect for appositives under a version of the direct-dissent diagnostic. Participants read declarative sentences introducing the target content and chose between two *no*-responses expressing direct dissent, targeting the appositive or the main-clause content (9).

(9) Syrett & Koev 2015, p. 542

A: *My friend Sophie, who performed a piece by Mozart, is a classical violinist.*
B1: *No, she's not.* (target: main clause)
B2: *No, she didn't.* (target: NRRC)

Direct dissent targeting the at-issue main clause (B1) is assumed to be always available. When the target content is not-at-issue, as with the NRRC content in (9), the B2 response is expected to be strongly dispreferred. Indeed, participants strongly preferred B1 responses (with proportions around 79%) when the competing alternative targeted a sentence-medial NRRC. Crucially, this preference was weaker for sentence-final NRRCs (around 65%), suggesting that final NRRCs may be interpreted as more at-issue than medial ones. Since this positional contrast for NRRCs has been argued by Snider (2017b) to be specific to assent and dissent-based diagnostics based on property (3c), a key question is whether it also emerges with other diagnostics. Therefore, one concrete way in which this paper will address research question (8a), whether different diagnostics yield consistent results, is testing whether they reproduce the positional effect for NRRCs observed in Syrett & Koev 2015.

Beyond NRRCs, Tonhauser et al. (2018) compared the 'asking whether' and 'are you sure?' diagnostics for contents associated with several English expressions, including sentence-medial NRRCs and complements of the clause-embedding predicates *annoyed, discover* and *know* (10).

(10) Tonhauser et al. (2018), p. 503
  a. annoyed: *Martha's neighbor is annoyed that Martha has a new BMW.*
  b. discover: *Mary discovered that her daughter has been biting her nails.*
  c. know: *Billy knows that Martha has a new BMW.*

The 'are you sure?' diagnostic presents participants with dialogues like (11) and collects ratings about whether the response addresses the question (see also the 'really' test of Shanon 1976).

(11) Tonhauser et al. (2018), p. 519
  Fred: *Martha's new car, a BMW, was expensive.*
  Carla: *Are you sure?*
  Fred: *Yes, I am sure that Martha's new car is a BMW.*

This diagnostic is taken to target property (3c), by testing whether the target content can be directly challenged. The target content is introduced by the initial assertion, the question challenges this assertion, and the response reinforces the target content. When the target content is at-issue, this response is expected to be judged as addressing the question, and as not addressing the question otherwise. For the nominal appositive in (11), the diagnostic is therefore expected to yield responses that indicate that the response is not addressing the question.

Tonhauser et al. (2018) found both convergence and divergence in their comparison (p. 526ff.). Ratings were correlated across diagnostics, suggesting that they may be sensitive to a shared underlying phenomenon. However, the 'asking whether' diagnostic yielded lower overall ratings and smaller differentiation between contents than the 'are you sure?' diagnostic.

Although the 'asking whether' diagnostic showed smaller between-content differentiation in Tonhauser et al. (2018), subsequent work shows that the 'asking whether' diagnostic can detect fine-grained differences. Degen & Tonhauser (2025: Exp. 1) found fine-grained at-issueness differences among the contents of the complements of 20 clause-embedding predicates in English. Fig. 1 shows the mean 'asking whether' ratings for the 20 predicates used in their study, ordered by mean rating.

Here, the complement of *annoyed* was rated as less at-issue than those of the other predicates tested, whereas that of *be right* received the highest ratings, and there were fine-grained by-predicate differences between many of the predicates in between.

Because the previous research indicates that diagnostics may differ in how strongly they differentiate among contents, a second way in which we concretely address research question i., about whether different diagnostics yield consistent results, is testing whether they reproduce fine-grained by-content differences for clause-embedding predicates observed in Degen & Tonhauser 2025.

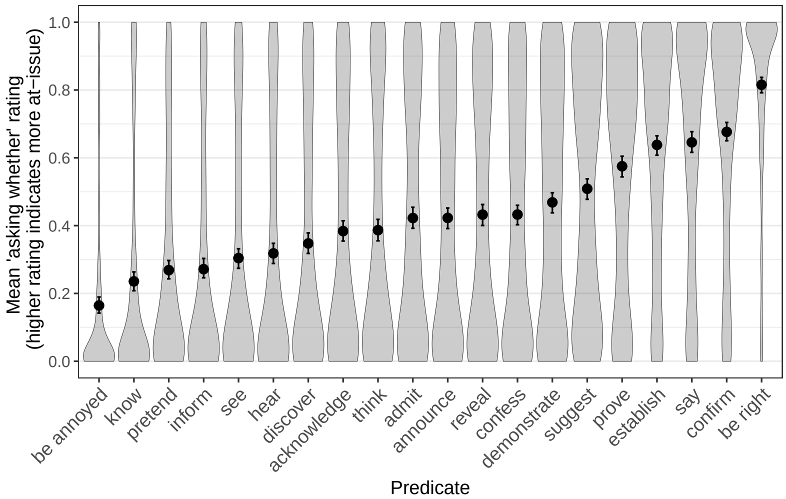


Figure 1: Mean 'asking whether' ratings for the contents of the clausal complements of 20 clause-embedding predicates, based on Degen & Tonhauser 2025: Exp. 1. Error bars indicate 95% bootstrapped confidence intervals. Violin plots indicate the the kernel probability density of individual participants' ratings.

1.5 Outlook

This paper presents a systematic experimental comparison of at-issueness diagnostics across six experiments, taking a first step towards addressing the two research questions in (8).

To address (8a), we compare diagnostics by testing whether they reproduce by-content contrasts reported in prior work, namely the medial-final contrast for NRRCs (Syrett & Koev 2015) and fine-grained differences among clause-embedding predicates (Degen & Tonhauser 2025). To begin addressing (8b), we consider how any diagnostic differences should be explained. In what follows, Section 2 reports on Experiments 1–4, which apply the four diagnostics in (4)–(7) to seven contents in English, including medial and final NRRCs and the complements of five clause-embedding predicates from Degen & Tonhauser (2025) (*confess, confirm, discover, know, be right*). None of these four experiments reproduced the positional effect for NRRCs reported by Syrett & Koev (2015). Only Exp. 2 ('asking whether') came close to observing the by-predicate differences found in Degen & Tonhauser (2025), with differences between *know, discover, confess, confirm, be right* and increasing mean at-issueness ratings in that order. The results suggest that diagnostics do vary in whether they detect at-issueness differences between particular expressions. Notably, the 'asking whether' diagnostic, which showed the greatest by-content differentiation in Exp. 2, is the only diagnostic which directly elicits judgments of speaker intentions, and the only one where the target content is embedded in a polar question.

Section 3 asks whether the the greater differentiation under the 'asking whether' diagnostic was due to the polar question embedding or the response task. Experiments 5 and 6 compared the 'asking whether' diagnostic with a direct-response diagnostic (Amaral et al. 2007; Tonhauser 2012) that also embeds content in a polar question but uses a different response task, eliciting acceptability judgments. Using the 20 clause-embedding predicates from Degen & Tonhauser (2025), the two experiments yielded highly correlated results, indicating that the observed high differentiation in Exp. 2 was driven by the polar question embedding, not the response task.

We argue in Section 4 that the speech act used to present the target content plays a central role: we propose that the illocutionary force of declarative sentences may interact with at-issueness ratings in a way that decreases by-content differentiation, while the illocutionary force of polar questions does impact the detection of differences between contents. Section 5 concludes.

2 Experiments 1-4

To compare the results of at-issueness diagnostics,[5] we conducted four experiments that each measured at-issueness with a different diagnostic, namely the QUD diagnostic (Exp. 1), the 'asking whether' diagnostic (Exp. 2), the direct-dissent diagnostic (Exp. 3) and the 'yes, but' diagnostic (Exp. 4).[6] Each experiment tested contents associated with the same set of seven of expressions in (12): the contents of sentence-medial and sentence-final NRRCs (12a–b), as well as the contents of the clausal complements of *be right, discover, confess, confirm, discover,* and *know* (12c–g).

(12) a. Content of sentence-medial NRRC
Lucy, who broke the plate, apologized. ⇝ Lucy broke the plate

b. Content of sentence-final NRRC
The police found Jack, who saw the murder. ⇝ Jack saw the murder

c. Content of the clausal complement of *know*
Ann knows that Raul cheated on his wife. ⇝ Raul cheated on his wife

d. Content of the clausal complement of *discover*
Mary discovered that Denny ate the last cupcake. ⇝ Denny ate the last cupcake

e. Content of the clausal complement of *be right*
Tom is right that Ann stole the money. ⇝ Ann stole the money

f. Content of the clausal complement of *confirm*
Harry confirmed that Greg bought a new car. ⇝ Greg bought a new car

g. Content of the clausal complement of *confess*
Lucy confessed that Dustin lost his key. ⇝ Dustin lost his keys

These seven contents were chosen because prior literature observed differences in at-issueness between two or more of these contents using a particular diagnostic for at-issueness. As discussed in §1, Syrett & Koev 2015 observed differences between sentence-medial and -final NRRCs using a variant of the direct-dissent diagnostic, and Degen & Tonhauser 2025 observed differences between *be right, discover, confess, confirm, discover,* and *know* using the 'asking whether' diagnostic. In particular, as shown in Fig. 1, the mean ratings of these five predicates span the range of ratings observed for the 20 predicates relatively evenly. Therefore, these five predicates provide a good test case for whether the diagnostics can detect fine-grained differences between contents. As a result, comparing these seven contents across the four diagnostics in Exps. 1–4 will allow us to assess whether the differences that emerge from one diagnostic also emerge from others. In each experiment, participants read the stimuli and gave ratings corresponding to the diagnostics.

Following prior work (e.g., Tonhauser et al. 2018), we describe higher/lower mean ratings by saying that content is more/less at-issue under a given diagnostic. However, we remain agnostic about whether at-issueness is an underlyingly binary or a gradient property (cf. Tonhauser et al. 2018; Barnes & Ebert 2023). On a gradient view, gradient mean ratings may be taken to reflect gradient at-issueness differences. On a categorical view, gradient mean ratings could be taken to reflect uncertainty about what the relevant QUD or proposal is, thus indicating the frequency or ease with which a particular QUD is attributed to the utterance.

[5] We use the term "at-issueness diagnostic" descriptively throughout the paper, even though our findings may ultimately suggest that some diagnostic tracks a different theoretical construct. We return to this issue in the general discussion.

[6] The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

2.1 Methods

2.1.1 Participants

For each of the four experiments, we recruited 80 unique participants on Prolific. These participants had registered on the platform as living in the USA and as having English as their primary language. They had at least 50 previous submissions and an approval rate of at least 97%. Table 1 shows the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 1 (QUD)	80	18-81 (43.8)	42/37/0/1
Exp. 2 (asking whether)	80	20-74 (38.5)	48/30/1/1
Exp. 3 (direct dissent)	80	18-77 (39.1)	50/28/1/1
Exp. 4 (yes, but)	80	19-67 (38.0)	48/30/2/0

Table 1: Information about the participants recruited in Exps. 1-4 (f = female, m = male, nb = nonbinary, dnd = did not disclose).

2.1.2 Materials and procedure

As shown in Fig. 2, the response options differed by diagnostic. In Exp. 1 (QUD diagnostic, panel (a)), participants rated how well the response fit the question on a slider marked 'totally doesn't fit' on one end (coded 0) and 'totally fits' on the other end (coded as 1). In Exp. 2 ('asking whether' diagnostic, panel (b)), participants judged whether the question was about the target content, using a slider marked 'no' on one end (coded as 0) and 'yes' on the other (coded as 1). In Exp. 3 (direct-dissent diagnostic, panel (c)), participants rated the naturalness of the direct dissent on a slider marked 'totally unnatural' (coded as 0) on one end and 'totally natural' on the other (coded as 1). Finally, in Exp. 4 ('yes, but' diagnostic, panel (d)), participants chose the response that sounded best; the two indirect dissents were coded as 0 and the direct one as 1. Across the four experiments, the responses were coded so that 1 meant that the content to be diagnosed was rated as at-issue and 0 as not-at-issue.

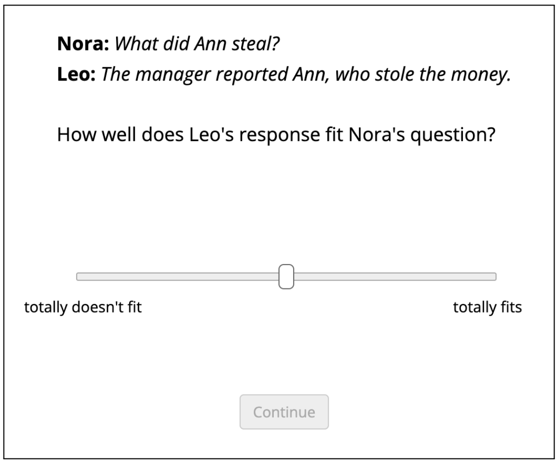
Each of the seven contents in (12) was instantiated by the seven items in (13) in each of the four experiments.

(13)
a. Jack saw the murder.
b. Raul cheated on his wife.
c. Ann stole the money.
d. Danny ate the last cupcake.
e. Lucy broke the plate.
f. Dustin lost his key.
g. Greg bought a new car.

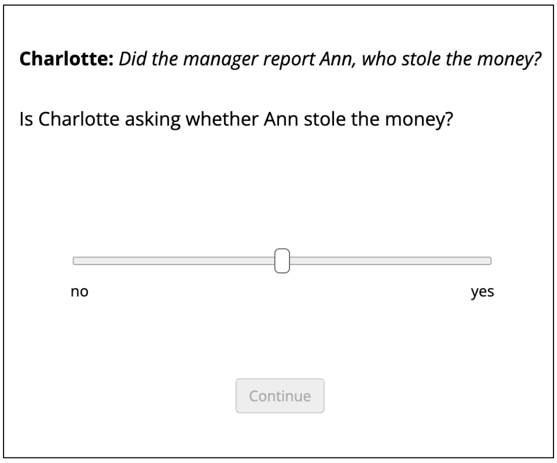
Each experiment included two control stimuli serving as attention checks: one was expected to receive a response at one end of the slider (Exps. 1-3) or a 'no' response (Exp. 4); the other control stimulus was expected to receive a response at the other end of the slider (Exps. 1-3) or a 'yes' response (Exp. 4). See Supplement A for the control stimuli used in Exps. 1-4.

In all four experiments, each participant's set of items was generated by randomly realizing each of the seven contents in (12) using a unique item in (13). Participants completed a total of 9 trials, namely 7 target trials and the same 2 control trials. Trial order was randomized for each participant.

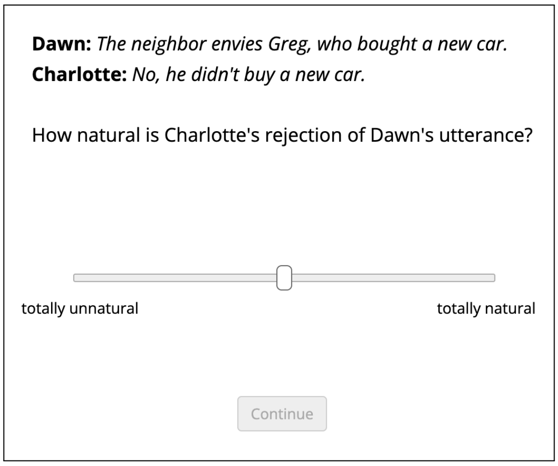
After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.



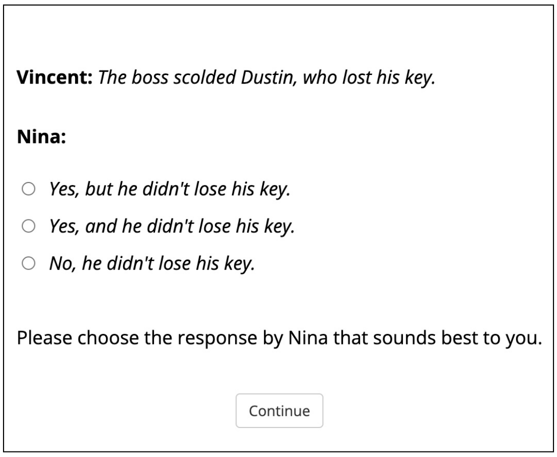
(a) Exp. 1: QUD diagnostic



(b) Exp. 2: 'asking whether' diagnostic



(c) Exp. 3: direct-dissent diagnostic



(d) Exp. 4: 'yes, but' diagnostic

Figure 2: Sample trials in (a) Exp. 1, (b) Exp. 2, (c) Exp. 3, and (d) Exp. 4.

2.1.3 Data exclusion

We excluded the data of participants that were not self-reported native speakers of American English and of participants whose responses to one of the two control trials was more than 2 sd away from the group mean (Exps. 1–3) or whose responses to one of the two control trials was incorrect (Exp. 4). Table 2 shows how many participants were excluded in each experiment, demographic information for the remaining participants, and the number of data points that entered into the analyses.

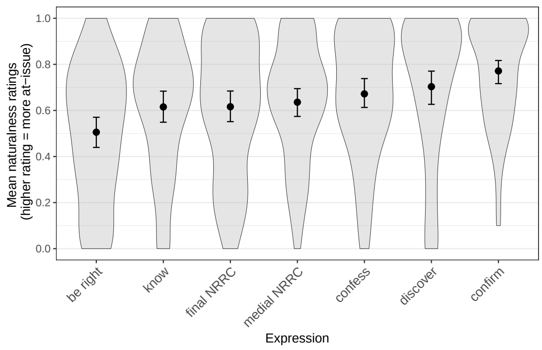
2.2 Results

Fig. 3 plots the results of the four experiments by the expression that is associated with the seven target contents. Panel (a) shows the mean naturalness ratings in Exp. 1 (QUD diagnostic), panel (b) mean 'asking whether' ratings in Exp. 2 ('asking whether' diagnostic), panel (c) shows mean

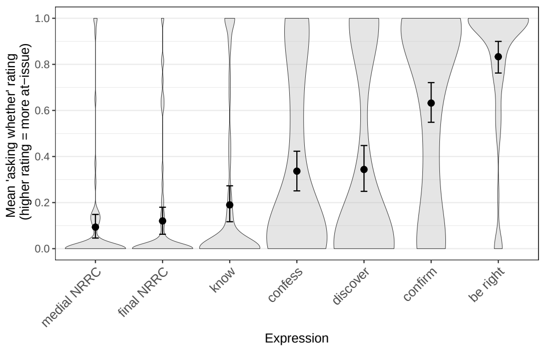
	exclusion criterion		remaining participants		data
	language	fillers	ages (mean age)	f/m/nb/dnd	points
Exp. 1 (QUD)	1	10	18-81 (41.1)	36/32/0/1	621
Exp. 2 (asking whether)	2	4	22-74 (38.7)	45/27/1/1	666
Exp. 3 (direct dissent)	2	7	18-77 (39.5)	44/25/1/1	639
Exp. 4 (yes, but)	4	4	19-67 (38.5)	43/27/2/0	648

Table 2: Information from Exps. 1-4 about the number of participants whose data was excluded based on their self-declared language (variety) and the fillers, about the remaining participants, and about the number of data points that entered into the analysis.

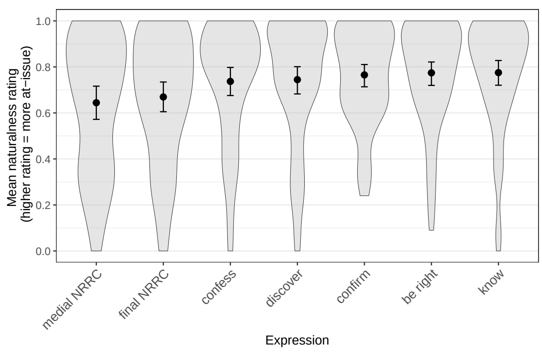
naturalness ratings in Exp. 3 (direct-dissent diagnostic) and panel (d) the proportion of 'no' choices in Exp. 4 ('yes, but' diagnostic).



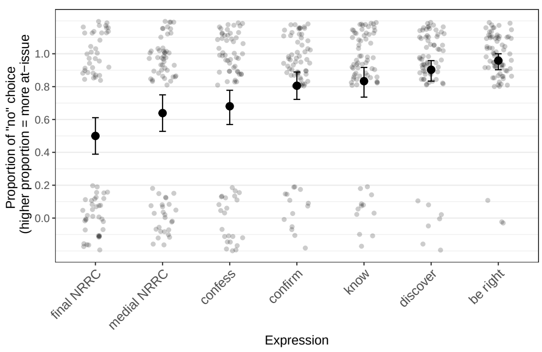
(a) Exp. 1 (QUD diagnostic)



(b) Exp. 2 ('asking whether' diagnostic)



(c) Exp. 3 (direct-dissent diagnostic)



(d) Exp. 4 ('yes, but' diagnostic)

Figure 3: Results of Exps. 1–4. Panels (a)–(c) show the mean responses by expression for (a) Exp. 1 (QUD diagnostic), (b) Exp. 2 ('asking whether' diagnostic), and (c) Exp. 3 (direct-dissent diagnostic); panel (d) shows the proportion of 'no' choices by expression in Exp. 4 ('yes, but' diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots in panels (a)-(c) show the kernel probability density of individual participants' ratings. Gray dots in panel (d) represent individual participant responses ('no' vs. 'yes', jittered vertically and horizontally for legibility).

As is visually apparent in Fig. 3, the diagnostics differ in the overall range of mean responses across the seven contents (given as the difference between the largest and smallest by-content means). The range is largest in Exp. 2 ('asking whether'), at .74 (from .1 to .83) and smallest in Exp. 3 (direct dissent), at .13 (.64 to .78). The results of Exp. 1 (QUD diagnostic), with a range of .27 (.51 to .77) and Exp. 4 ('yes, but'), with a range of .46 (.5 to .96), fall in between.

The experiments differ in whether they reproduced distinctions between contents reported in prior work. Syrett & Koev (2015), using a variant of the direct-dissent diagnostic, found that sentence-final NRRCs are more at-issue than sentence-medial ones. In contrast, as shown in Fig. 3, no such difference was observed in Exps. 1–3; and in Exp. 4 ('yes, but'), we found the opposite: sentence-final NRRCs were rated less at-issue than sentence-medial ones. Thus, none of the diagnostics as implemented in Exps. 1–4 replicate Syrett & Koev's finding.

For clause-embedding predicates, Exp. 2 ('asking whether') largely reproduced the pattern from Degen & Tonhauser (2025), namely, that complements of *know* were rated as less at-issue than those of *discover*, followed by *confess*, then *confirm*, and finally *be right*, with the only exception that *confess* and *discover* did not differ here. The other three experiments showed fewer differences between the mean ratings or proportions among the clause-embedding predicates. In Exp. 1 (QUD diagnostic), the embedded content of most predicates showed little differentiation, with only two predicates showing a clearly different pattern: the embedded content of *confirm* was rated most at issue, and the embedded content of *be right* was rated less at-issue than the other predicates. The direction of this difference is opposite to that observed in prior literature and the other experiments. Exp. 3 showed no clear differences among clause-embedding predicates. In Exp. 4, the proportions of *yes*-responses were similar for many of the clause-embedding predicates, but the highest proportion was found for content of the complement of *be right*, which was clearly different from the predicates on the lower end, such as *confess, confirm* and *know*.

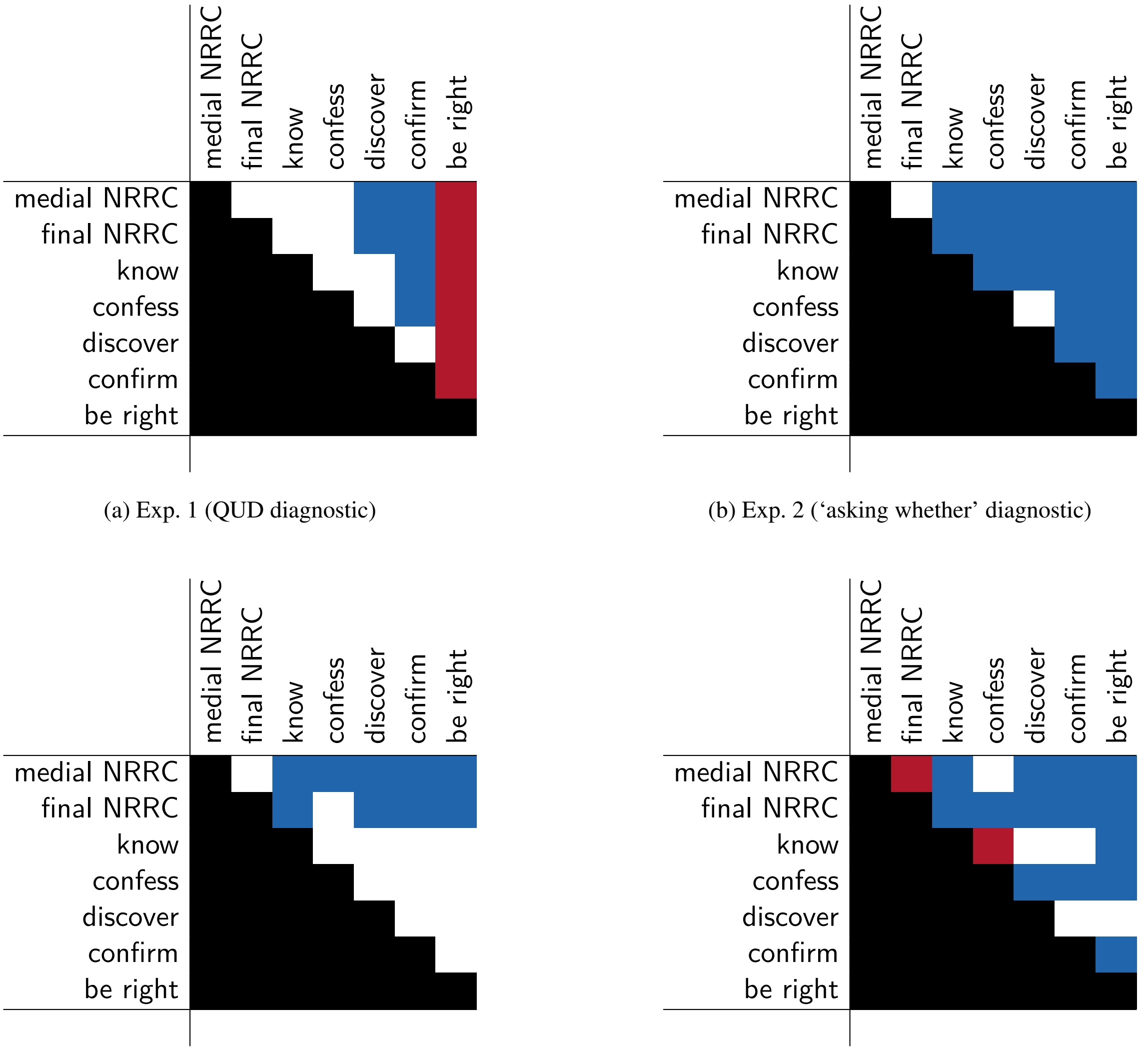
The differences described above are supported by post-hoc pairwise comparisons of the estimated means or proportions for each content, using the `emmeans` package (Lenth 2023) in R (R Core Team 2016), illustrated in Fig. 4. These comparisons were based on mixed-effects beta regression models (Exps. 1–3) and a mixed-effects logistic regression model (Exp. 4), all fit using the `brms` package (Bürkner 2017) using weakly informative priors. The models predicted the ratings[7] from a fixed effect of expression (treatment-coded, with *be right* as reference level), with random intercepts for participants and items. The output of the pairwise comparison were 95% highest density intervals (HDIs) of estimated marginal mean differences between each of the expressions. We assume that two contents differ if their HDI does not include 0.[8]

Consistent with the patterns described above, Fig. 4 shows that the only positional difference for NRRCs is observed in Exp. 4 ('yes, but'), where sentence-medial ones are rated more at-issue than sentence-final ones, opposite to the finding in Syrett & Koev (2015). Among clause-embedding predicates, Exp. 2 ('asking whether') shows the most pairwise differences, while Exp. 3 (direct dissent) shows none.

We also find that the four experiments differ in the relative ranking of the seven contents, with only limited overlap. Across all experiments, *confirm* and *discover* consistently received higher ratings (at least numerically) than *confess*, which in turn received higher ratings than medial and final NRRCs. For all other pairs of expressions, however, the ordering was not consistent. In particular, the ranking of *be right* relative to most other contents is inconsistent across experiments: The embedded content of *be right* received the lowest ratings in Exp. 1, but was among the most

[7] To model the ratings in Exps. 1–3 using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.

[8] The full model outputs are available in the folder `results+analysis/exps1-4/` in the repository available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.



(a) Exp. 1 (QUD diagnostic)

(b) Exp. 2 ('asking whether' diagnostic)

(c) Exp. 3 (direct-dissent diagnostic)

(d) Exp. 4 ('yes, but' diagnostic)

Figure 4: Pairwise differences between expressions, ordered from top to bottom and left to right by increasing mean in Exp. 2 ('asking whether' diagnostic). A white cell means that the 95% HDI of the pair of the row expression and the column expression includes 0, a red cell means that the 95% HDI does not include 0 and that the coefficient is positive (the row expression received a higher rating than the column expression), and a blue cell means that the 95% HDI does not include 0 and the coefficient is negative (the row expression received a lower rating than the column expression).

at-issue in the other three experiments. This difference between the results of the experiments is quantified in the Spearman rank correlations shown in Table 3.[9]

[9] The Spearman rank correlation coefficient, a value between -1 and 1, is a nonparametric measure of rank correlation: the higher the absolute value of the coefficient, the more the relation between the two variables can be described using a monotonic function. If the coefficient is positive, the value of one variable tends to increase with an increase in the other. In the case of our experiments, a coefficient of 1 for two experiments would mean that there is a perfectly monotone increasing relation between the mean ratings of the seven contents in the two experiments: for any two contents c1 and c2, if c1 ranks below c2 in one experiment (that is, the mean rating of c1 is lower than that of c2), then that ranking is preserved in the other experiment.

	Exp. 1	Exp. 2	Exp. 3	Exp. 4
Exp. 1 (QUD diagnostic)		.11	-.29	-.18
Exp. 2 (‘asking whether’ diagnostic)			.64	.79
Exp. 3 (direct-dissent diagnostic)				.79

Table 3: Spearman rank correlations between the results of Exps. 1–4.

The rank correlations are particularly low for Exp. 1 compared to the other three experiments, at least partly because of the low relative ranking of *be right*. These results suggest that the four diagnostics as implemented in Exps. 1–4 interact differently with the seven contents investigated.

2.3 Discussion

Exps. 1–4 were designed to compare four at-issueness diagnostics that have been widely used in the literature. To assess whether the diagnostics yield consistent findings when applied to the same target contents, we asked whether they reproduce contrasts reported in prior work, particularly positional effects for NRRCs, and fine-grained differences among the contents of the complements of five clause-embedding predicates. The results of Exps. 1–4 suggest that the four diagnostics differ in whether they reproduce these contrasts, and in the relative order of the seven contents investigated. Three main differences between the four experiments emerged:

i. Although none of the diagnostics as implemented here reproduce the positional effect for NRRCs reported by Syrett & Koev (2015), only Exp. 4 (‘yes, but’) found a difference for medial and final NRRCs. However, this was the opposite pattern from Syrett and Koev, namely that medial NRRCs were judged as more at-issue than final ones. Exps. 1–3 found no significant difference between medial and sentence-final NRRCs.
ii. The results of the diagnostics differ in the relative order of the seven contents—most notably in the case of the complement of *be right*, which ranks low under the QUD diagnostic but high under the other diagnostics.
iii. The four diagnostics differ in the extent to which they differentiate between the contents of complements of the five clause-embedding predicates as in Degen & Tonhauser (2025): Only the ‘asking whether’ diagnostic (Exp. 2) comes close to producing the differences.

This section discusses the implications of these findings for the interpretation of the diagnostics and for theories of at-issueness.

2.3.1 Positional effects for NRRCs?

Syrett & Koev 2015: Exp. 2 found that the content of sentence-final NRRCs was relatively more at-issue than that of sentence-medial ones, using a direct-dissent diagnostic in which participants chose between directly dissenting either with the NRRC content or with the assertion contributed by the declarative main clause. None of our Exps. 1–4 replicated this finding: Exps. 1–3 found no positional difference, while Exp. 4 (‘yes, but’) showed the opposite pattern, where sentence-medial NRRCs were rated as more at-issue than sentence-final ones.

While Exp. 2 (asking whether’) found no positional effect for NRRCs, it otherwise revealed substantial differentiation among the complements of the five clause-embedding predicates investigated. We hypothesize that the absence of a positional effect may be due to an interaction between the diagnostic and the illocutionary properties of NRRCs. The ‘asking whether’ diagnostic presents target contents in polar questions and asks participants whether the speaker is asking about the target content. At the same time, appositive NRRCs have been argued to contribute a speech act that is separate from the speech act conveyed by their host sentence (see, e.g., Potts 2005; Koev

2013; Syrett & Koev 2015; **?**). **LH: I'm not sure about how to cite the Krifka paper. As far as I can see, the published version does not contain the part about NRRCs any more.**

Under this view, the sample stimuli in (14) convey two distinct speech acts: the content of the NRRC is an assertion and the main clause content is an (information-seeking) question.

(14) Sentence-medial NRRC: *Does Greg, who bought a new car, have a driver's license?*
Sentence-final NRRC: *Does the neighbor envy Greg, who bought a new car?*

Question for participants: Is the speaker asking whether Greg bought a car?

If the appositive NRRC contributes a separate assertion rather than part of the question, participants would not be expected to judge the speaker as asking about its content. This is consistent with the very low mean ratings for both medial and sentence-final NRRCs in Exp. 2. In contrast, the declarative stimuli in Exps. 1, 3, and 4 present both the main-clause content and the appositive as assertions, making them more similar to the declarative stimuli used by Syrett & Koev (2015).

In discussing their findings, Syrett and Koev propose that the appositive assertion competes with the main-clause assertion for at-issue status. If this analysis is correct, and if such competition requires the competing contents to have the same illocutionary force, then the absence of a positional effect in Exp. 2 is expected: an appositive that contributes an assertion may not be in a position to compete for at-issueness status with an interrogative main clause. Since Syrett and Koev attribute the positional effect to the influence of linear order on this competition, this kind of explanation could provide an understanding of why the positional effect is absent at least in Exp. 2, where the target content was presented in an interrogative main clause.

In the other three experiments, target contents were presented in declarative main clauses, which contribute assertions. If Syrett and Koev's analysis is correct, appositive NRRCs should therefore, in principle, be able to compete with the main-clause assertion for at-issue status. Among our diagnostics, the dissent-based tasks in Exps. 3 (direct dissent) and Exp. 4 ('yes, but') most closely resemble the diagnostic used in Exp. 2 of Syrett & Koev (2015), but neither of these replicated their finding: Exp. 3 showed no positional effect, whereas Exp. 4 produced the opposite pattern.

One possible source of this divergence is the response task. Although all three studies used dissent-based diagnostics with target content presented in declarative assertions, Syrett & Koev's 2015 forced-choice implementation implementation differed from ours. Exp. 3 elicited naturalness ratings rather than forced-choice responses. Exp. 4 did use a forced-choice continuation task, but its response options differed substantially from those used by Syrett & Koev. In our implementation (based on Xue & Onea 2011), all response options rejected the target content, and participants chose between expressing direct dissent (*no*), or indirect dissent (with *yes, but* or *yes, and*). In contrast, in Syrett & Koev's study, all response options expressed direct dissent (with *no*), and participants chose whether to reject the NRRC content or the main clause content.

Taken together, these results suggest that positional effects for NRRCs may not be robust across diagnostics and, furthermore, the corresponding task differences suggest that even small changes in the response task used to operationalize a diagnostic, such forced-choice alternatives, could reverse the observed pattern. More generally, different diagnostics vary in whether they detect any positional difference for NRRCs at all. This suggests that previously reported effects may reflect task- or diagnostic-specific effects rather than stable properties of NRRC interpretation. There are a number of differences between the three diagnostics that may have contributed to these different results: the stimuli, the response task, the response options. It is a pressing question for future research to identify whether the results obtained in the three experiments replicate and, if yes, why these three diagnostics give distinct results. A natural follow-up would be to test whether applying their particular forced-choice schema to our stimuli would reproduce their effect.

2.3.2 Interactions with discourse requirements: the case of *be right*

The diagnostics, as implemented here, yield different rank orderings of contents. The most pronounced ranking difference concerns the complement of *be right*: it receives the lowest ratings in Exp. 1 (QUD diagnostic) and relatively high ratings in the other experiments. At first glance, this divergence appears to indicate that the complement of *be right* is not at-issue under the QUD diagnostic but is at-issue under the other diagnostics. However, we argue that the low mean rating for the complement of *be right* is better explained by how the QUD diagnostic interacts with the discourse meaning conveyed by *be right*.

Consider that, as shown in panel (a) of Fig. 3, participants gave relatively low naturalness ratings to responses such as (15a), indicating that they did not judge the response as fitting the question.

(15) Sample stimuli from the four experiments with *be right*

a. Exp. 1 (QUD diagnostic)
Nora: *What did Lucy break?*
Leo: *Danny is right that she broke the plate.*

b. Exp. 2 ('asking whether' diagnostic)
Nora: *Is Danny right that Lucy broke the plate?*

c. Exp. 3 ('direct dissent' diagnostic)
Nora: *Danny is right that Lucy broke the plate.*
Leo: *No, she didn't break the plate.*

d. Exp. 4 ('yes, but' diagnostic)
Nora: *Danny is right that Lucy broke the plate.*
Leo: *Yes, but she didn't break the plate.*
Yes, and she didn't break the plate.
No, she didn't break the plate.

We hypothesize that the low ratings in Exp. 1 arise because utterances which include *be right* in a veridical environment, like Leo's utterance in (15a), convey that the attitude holder (here, Danny) has publicly committed to the content of the clausal complement (here, that Lucy broke the plate),[10] as made explicit in (16).

(16) a. **Danny:** *Lucy broke the plate.*
b. **Nora:** *What did Lucy break?*
c. **Leo:** *Danny is right that she broke the plate.*

We can understand the ill-formedness of (16) with a QUD-based framework of discourse structure (in the sense of Ginzburg 1996; Roberts 2012; Büring 2003). Within this framework, Danny's assertion is assumed to address the question whether Lucy broke the plate.[11] The inference associated with (16c), that Danny has previously committed to Lucy breaking the plate, therefore can be interpreted within a QUD-framework as conveying that the question "whether Lucy broke the plate" is on the QUD-stack of previously raised questions at the point where *be right* is uttered.

[10] Abusch (2002; 2010) characterizes *be right* as conveying a projective doxastic inference, namely that the subject believes the embedded proposition. Following a suggestion in Schlenker 2010 (p. 135), Anand & Hacquard (2014: p. 73) argue that the content of this projective inference instead reports an assertive speech act, conveying that the subject has committed to the embedded content by means of a prior discourse move.

[11] While asserting p is typically assumed to raise the issue of whether p (see, e.g., Ginzburg & Sag 2000; Farkas & Bruce 2010), this may well be a subquestion (i.e., part of an answering strategy in the sense of Roberts 2012) of a superquestion such as "What did Lucy break?" or *Who broke the plate?*, since an answer to *whether p* entails a partial answer to either of the latter two (see definition of subquestion in Roberts 2012, p.6:16).

This QUD is a subquestion of Nora's question in (16b) "What did Lucy break", since every complete answer to the latter entails an answer to the former. The progression from the subquestion to the explicitly given superquestion violates Roberts's 2012 constraint on QUD stacks (p. 6:15), which allows only the reverse order (from superquestions to subquestions).[12]

As a result, the low ratings in Exp. 1 need not indicate that the embedded proposition is not at-issue but may be due to an infelicitous response in the given discourse context.[13] In the other diagnostics (15b)–(15d), where no prior discourse is given, the ratings for *be right* are not affected by its inference about a previous discourse commitment. This contrast highlights that diagnostics may interact with literal meaning independently of at-issueness, and such interactions must be taken into account when interpreting the results.

2.3.3 Why does the 'asking whether' diagnostic show greater differentiation?

A striking difference between the results of the experiments is that Exp. 2 ('asking whether') showed the greatest differentiation between contents, whereas the other three experiments exhibited a smaller range of means, and fewer statistically supported differences between investigated contents. Among the first four experiments, Exp. 2 was the only one that came close to reproducing the fine-grained differences among the complements of the five clause-embedding predicates that were differentiated in Degen & Tonhauser (2025).

One question we can consider here is whether this difference between the diagnostics as implemented in Exps. 1–4 could be explained by assuming a split between diagnostics, targeting empirically distinct discourse phenomena captured by the definitions in terms of QUD-at-issueness and proposal at-issueness. The asking whether' diagnostic targets property (3b), according to which the at-issue content of a question determines the relevant set of alternatives. Snider (2017b) has linked this property to the QUD-based definition of at-issueness.[14] Consequently, if the results had produced diagnostic differences where Exp. 2 ('asking whether') patterns together with Exp. 1 (QUD-diagnostic) and differently from Exp. 3 (direct dissent) and Exp. 4 ('yes but'), they could potentially have been explained by assuming that the diagnostics track distinct underlying phenomena. However, since Exp. 2 uniquely differed from the other experiments, we need to consider other explanations.

One salient difference between the 'asking whether' diagnostic (Exp. 2) and the other diagnostics (Exps. 1, 3, and 4) is the clause type in which the target content is presented. In Exp. 2, target contents are embedded in interrogatives, which are typically interpreted as information-seeking questions, whereas in the other experiments they are embedded in declaratives, which are interpreted as assertions. We hypothesize that this difference in clause type, and associated illocutionary force differences, may have affected the degree of by-content differentiation.

Recall that the five clause-embedding predicates were selected because their complements differed substantially under the 'asking whether' diagnostic in Degen & Tonhauser 2025. In declaratives,

[12] This constraint on QUD stacks can be understood as an informativeness-constraint, which requires that discourse moves contribute towards increasing the information quantity in the context set. Accordingly, the exchange in (16) is acceptable only under an echo-question interpretation of Nora's question.

[13] When *be right* is excluded, the Spearman rank correlations are:

	Exp. 1	Exp. 2	Exp. 3	Exp. 4
Exp. 1 (QUD diagnostic)		.77	-.09	-.31
Exp. 2 ('asking whether' diagnostic)			.66	.66
Exp. 3 ('direct dissent' diagnostic)				.77

[14] Specifically, Snider 2017b suggests that Tonhauser's 2012 property (3b) can be understood as a forward-looking counterpart of property (3a). While property (3a) concerns addressing an established QUD, property (3b) concerns determining the alternatives an utterance introduces. Snider's suggestion that property (3b) aligns with the QUD-based definition can be substantiated by the assumption that the alternatives introduced by an utterance are the possible answers to the QUD raised by the utterance (Roberts 2012; Büring 2003; Beaver & Clark 2008).

however, the complements of these predicates exhibit a rather uniform profile. Specifically, Degen & Tonhauser 2022: Exps. 2 investigated the extent to which the CCs of the same 20 clause-embedding predicates (and same 20 items) as in Degen & Tonhauser 2025 are inferred from utterances of declaratives. The contents of clausal complements of the five clause-embedding predicates investigated in our Exps. 1–4 all received very high mean inference ratings on a scale from 0–1 (with the lowest being .9 for *confess*). What these results mean is that an interpreter who utters a declarative with one of these five clause-embedding predicates in one of our Exps. 1, 3, 4 is very likely to infer that the complement clause is true.

We hypothesize that this inference to the truth of the complement may have contributed to the lesser differentiation among the five clause-embedding predicates in Exps. 1, 3, 4. When a participant reads a speaker's utterance of a declarative with one of the five predicates in Exps. 1, 3, 4, such as *confirmed* in (17), they are likely to attribute commitment to the content of the complement clause p to the speaker. After all, if the speaker had not intended to convey such a commitment, they could have chosen a different predicate.

(17) *Helen confirmed that Tony had a drink last night.*

a. p: *Tony had a drink last night* (target content)

b. q: *Helen confirmed* p (main-clause content)

In Exps. 1, 3, and 4, participants are therefore likely to take the speaker to be committed both to the main-clause content (here: q) and to the complement clause content (here: p). and the target content shares an important property with the at-issue main-clause content: both are treated as speaker commitments.

The hypothesis that participants take the speaker to be committed to both the content of the main clause and the complement clause could help explain the patterns of differences between diagnostics observed in Exps. 1–4. In Exp. 1 (QUD diagnostic), participants rated the naturalness of a declarative response to a question addressed by the complement clause. The relatively high ratings for *know*, *confess*, *discover*, and *confirm* may reflect the fact that the speaker is taken to be committed to the complement clause, making it a plausible answer to the preceding question. We leave *be right* aside here for the reasons discussed above. In, Exp. 3 (direct dissent) and Exp. 4 ('yes, but'), participants rated the naturalness of a direct dissent of the complement or chose between direct and indirect dissents to the target content. If these diagnostics are sensitive to the availability of the CC for anaphoric reference, the hypothesis that the speaker is committed to each of these CCs may also mean that each of the CCs is available for anaphoric reference. **LH: Since we can easily refer back to content that the speaker is not committed to, I would rather relate this to a different possible explanation. I am thinking about Syrett and Koev's suggestion that asserted content may compete for at-issueness status.**

Another difference between the 'asking whether' diagnostic (Exp. 2) and the other diagnostics (Exps. 1,3,4) is that the 'asking whether' diagnostic asks participants to judge the speaker's intended content of the main speech act whereas the other three diagnostics ask participants to rate the naturalness of a dialogue (Exp. 1), the naturalness of a direct dissent (Exp. 3) or choose between direct and indirect dissent options (Exp. 4). One possibility is that the response task is what matters. This is why we ran Exps. 5 and 6. To tease apart these two possible explanations for the greater by-content differentiation of the 'asking whether' diagnostic, the next section compares the 'asking whether' diagnostic (Exp. 5) to a diagnostic that also embeds the target content in a polar question but elicits naturalness ratings on direct responses (Exp. 6).

3 Experiments 5 and 6

Exps. 5 and 6 were designed to investigate whether the greater differentiation observed in Exp. 2 (‘asking whether’) compared to Exps. 1, 3 and 4 was due to (i) embedding target contents in polar questions or (ii) the response task. Together with the results of Exps. 1–4, the results of Exps. 5–6 allow us to test two hypotheses:

(18) Hypotheses

a. **Question-embedding hypothesis:** Differentiation between contents is greater when contents are embedded in a polar question than in a declarative sentence.

b. **Response-task hypothesis:** Differentiation between contents is greater when participants are asked directly what the utterance is about, compared to tasks such as naturalness judgments or forced-choice continuations.

Exps. 5 and 6 compared two diagnostics that both embed the target content in a polar question but differ in the response task. Exp. 5 used the ‘asking whether’ diagnostic as in Exp. 2 (19a) and Exp. 6 uses a direct-response diagnostic (Amaral et al. 2007; Tonhauser 2012), shown in (19b).

(19) Implementation of the diagnostics in Exps. 5–6

a. Exp. 5 (‘asking whether’ diagnostic)
Nora: *Is Tom right that Lucy broke the plate?*
Question to participants: Is Nora asking whether Lucy broke the plate?

b. Exp. 6 (direct-response diagnostic)
Nora: *Is Tom right that Lucy broke the plate?*
Leo: *Yes, Lucy broke the plate.*
Question to participants: Does Leo’s response to Nora sound good?

The direct-response diagnostic is based on property (3b), that is, that the at-issue content of a polar question determines the question alternatives. On this diagnostic, participants read a dialogue in which one speaker asks a polar question containing the target content, and then rate the naturalness of a *yes* response that asserts the target content. Such a response is natural only if the target content provides a possible answer, that is, a question alternative (von Stechow 1991). Accordingly, participants are expected to judge a dialogue as natural when the target content is provides a question alternative and as unnatural otherwise.

Under the question-embedding hypothesis in (18a), the greater differentiation in Exp. 2 (‘asking whether’) is attributed to presenting the target content in a polar question rather than a declarative. Thus, under this hypothesis, Exps. 5 and 6 should both show high differentiation. On the other hand, under the response-task hypothesis in (18b), the greater differentiation in Exp. 2 is attributed to probing speaker intentions rather than collecting metalinguistic judgments such as naturalness ratings. Under this hypothesis, Exp. 5 (‘asking whether’) should show high differentiation, whereas Exp. 6 (direct response) should not, since it uses naturalness judgments, like Exps. 1, 3, and 4.

Both experiments measured at-issueness of the contents of the complements of the 20 clause-embedding predicates from Degen & Tonhauser 2025, which were shown in Fig. 1.

(20) 20 clause-embedding predicates from Degen & Tonhauser 2025:

acknowledge, admit, announce, be annoyed, be right, confess, confirm, demonstrate, discover, establish, hear, inform, know, pretend, prove, reveal, say, see, suggest, think

Both experiments also assessed the projection of the complement contents. This data was reported in Hofmann et al. 2024. Here we focus on the at-issueness data, which have not yet been reported.

3.1 Methods

3.1.1 Participants

We recruited 284 participants on Amazon's Mechanical Turk platform for Exp. 5 and 250 participants on Prolific for Exp. 6.[15] Participants recruited on Mechanical Turk had U.S. IP addresses and at least 99% of previous HITs approved. Participants recruited on Prolific had registered as U.S.-born native speakers of English residing in the USA and had an approval rate of at least 99%. Table 4 summarizes the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 5 (asking whether)	284	19-74 (38.2)	–/–/–/–
Exp. 6 (direct response)	250	18-58 (25.5)	201/43/6/0

Table 4: Information about the participants recruited in Exps. 5-6 (f = female, m = male, nb = nonbinary, dnd = did not disclose; gender information was not collected in Exp. 5).

3.1.2 Materials and procedure

Target stimuli consisted of a polar question uttered by a named speaker in Exp. 5, as shown in (19a), and of a dialogue consisting of a polar question and a response uttered by a named speaker, respectively, in Exp. 6, as shown in (19b). Target stimuli in both experiments were constructed by combining each of the 20 clause-embedding predicates in (20) with one of 20 items (see Supplement B). Both experiments included the same six control stimuli, where target contents were expected to be at-issue (see Supplement C).

As shown in Fig. 5, while both experiments featured a polar question where the target content was realized as the content of a clause-embedding predicate, the response task differed between the experiments. In Exp. 5 ('asking whether', panel (a)), participants judged whether the question was about the content of the complement, using a slider marked 'no' on one end (coded as 0) and 'yes' on the other (coded as 1). In Exp. 6 (direct response, panel (b)), participants rated whether the response to the first speaker sounds good, on a slider marked 'no' (coded as 0) and 'yes' (coded as 1) on the two ends. In the initial instructions, participants were asked to imagine they are at a party and that, when walking into the kitchen, they overhear somebody say something to somebody else. In both experiments, the responses were coded so that 1 meant that the target content was rated as at-issue and 0 as not-at-issue.

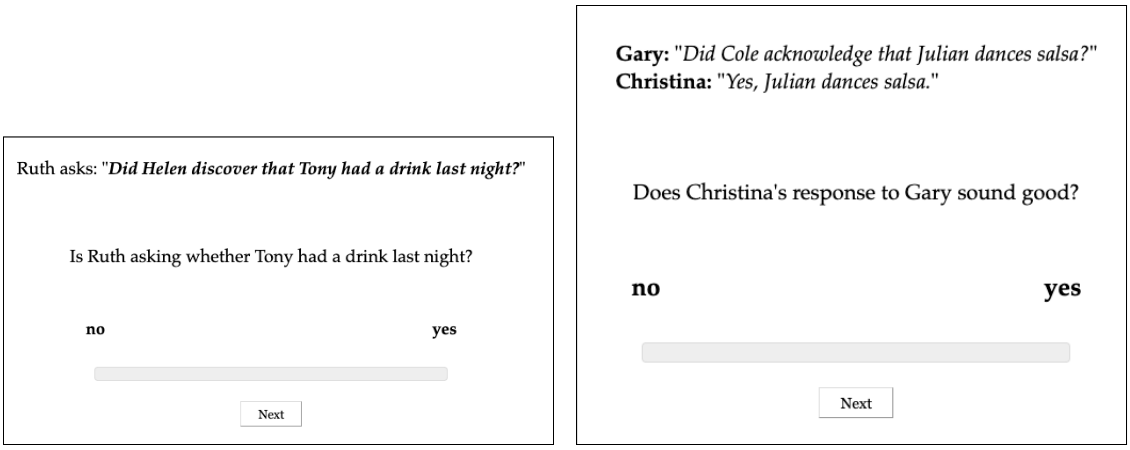
For each participant, the 20 predicates were randomly paired with the 20 items, yielding one target stimulus per predicate. Participants thus saw 26 stimuli in total: 20 target items and 6 controls.[16] Trial order was randomized for each participant. After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

3.1.3 Data exclusion

We excluded data from participants who did not self-identify as native speakers of American English, and those whose responses to one of the two control trials was more than 2 sd from the group mean. Table 5 shows the number of exclusions in each experiment, the characteristics of the remaining participants, and the number of data points that entered into the analyses.

[15] Exp. 5 was run in August 2019 and Exp. 6 in August 2021.

[16] Each participant saw their set of 26 stimuli twice, once in the projection block and once in the at-issueness block. Block order was randomized. As mentioned above, we focus here on the at-issueness data.



(a) Exp. 5: 'asking whether' diagnostic

(b) Exp. 6: direct response diagnostic

Figure 5: Sample trials in (a) Exp. 5 and (b) Exp. 6.

	exclusion criterion		**remaining participants**			data
	language	controls	ages (mean age)	f/m/nb/dnd	total	points
Exp. 5 (asking whether)	7	35	21-74 (39.2)	–/–/–/–	242	6292
Exp. 6 (direct response)	5	24	18-58 (24.9)	187/29/5/0	221	5746

Table 5: Information from Exps. 5-6 about the number of participants whose data was excluded based on their self-declared language and language variety ('language'), the controls, and the variance of their responses, about the remaining participants, and about the number of at-issueness data points that entered into the analysis.

3.2 Results and discussion

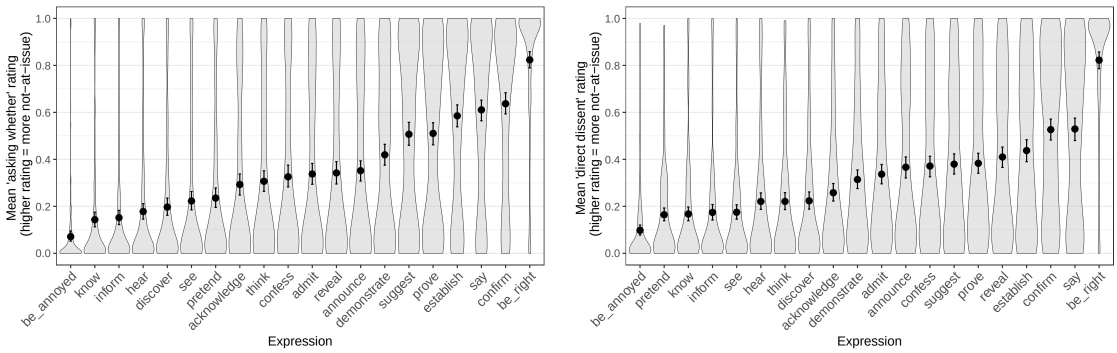
Fig. 6 shows the results of the two experiments by predicate: panel (a) plots mean 'asking whether' ratings in Exp. 5, and panel (b) plots mean naturalness ratings in Exp. 6 (direct-response).

Overall, the results of two experiments patterned similarly. The ranges of by-content means are comparable (Exp. 5 range: .75, .07 to .82; Exp. 6 range: .73, .09 to .82), and the relative orderings of contents by their means align closely, as reflected in a strong Spearman rank correlation ($r_s = .93$).

Importantly, both experiments show fine-grained by-content differentiation. For instance, for the subset of predicates also tested in Exps. 1–4, Exps. 5 and 6 reproduced the ordering found in Degen & Tonhauser (2025) and show pairwise differences in the predicted directions for adjacent contents, with increasing mean ratings between *know, discover, confess, confirm, be right* in that order.

There are also some differences between the experiments, concerning a small number of local reversals in the middle of the scale. For example, the content of the complement of *demonstrate* came out as more at-issue than that of *announce*, *confess*, and *reveal* under the 'asking whether' diagnostic as implemented in Exp. 5, but as less at-issue than the content embedded under these predicates under the implementation of the direct-response diagnostic in Exp. 6.

Fig. 7 presents the results of post-hoc pairwise comparisons of the estimated means for each content. As in § 2, we used `emmeans` (Lenth 2023) in R (R Core Team 2016) applied to mixed-effects beta regression models fit with `brms` (Bürkner 2017) using weakly informative priors.



(a) Exp. 5 ('asking whether' diagnostic)

(b) Exp. 6 ('direct assent' diagnostic)

Figure 6: Results of Exps. 5–6. The panels show the mean ratings by expression for (a) Exp. 5 (asking whether diagnostic) and (b) Exp. 6 (direct response diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots show the kernel probability density of individual participants' ratings.

The models predicted ratings[17] from a fixed effect of content (treatment-coded, with *be right* as the reference level). The pairwise comparisons support the above generalizations: while the two diagnostics as implemented here differ in which specific contrasts they detect, their overall patterns are highly consistent. Both diagnostics reveal fine-grained at-issueness differences between contents. At the same time, there are some differences, namely some contrasts that reached significance in experiment did not in the other.

In sum, the two diagnostics as implemented Exps. 5 and 6 ('asking whether' and direct-response) yielded highly similar results for the 20 contents investigated, including fine-grained by-predicate differentiation, even though they differed in response task. This result supports the question-embedding hypothesis in (18a) over the response-task hypothesis in (18b).

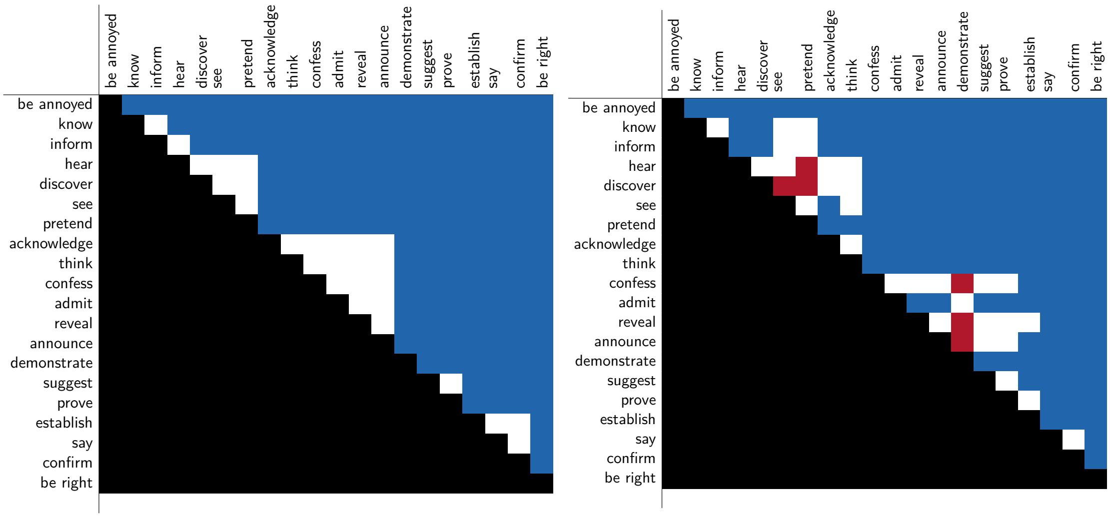
Nonetheless, small differences between the results of Exps. 5 and 6 remain, for example in which by-content differences reached significance and in the precise rank ordering of contents. Since the two experiments used identical embedding contexts and both diagnostics are based on property (3b), which has been linked to a QUD-based definition (Snider 2017b), these differences likely stem from the difference in response task, suggesting again that response task differences can affect at-issueness judgments in subtle ways.

4 General discussion

This paper reported six experiments that were designed to address the two research questions in (8), namely, (8a) whether different diagnostics yield consistent results when applied to the same contents, and (8b) whether differences between diagnostics reflect distinct underlying notions of at-issueness.

The results of the experiments provide a clear answer to the research question in (8a): the five diagnostics did not all yield consistent results. Although the results of Exp. 5 ('asking whether') and Exp. 6 (direct response) gave largely consistent results when applied to the contents of the clausal complements of 20 clause-embedding predicates, the results of Exps. 1–4 differed in the

[17] To model the ratings using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.



(a) Exp. 5 (‘asking whether’ diagnostic)

(b) Exp. 6 (direct-response diagnostic)

Figure 7: Pairwise differences between expressions, ordered from by increasing mean in Exp. 5 (‘asking whether’). White cells indicate that the 95% HDI of the difference includes 0. Red cells indicate a positive difference (the row expression received a higher rating than the column expression), and blue cells indicate a negative difference.

extent to which they replicated previously reported differences among the contents of NRRCs and the complements of five clause-embedding predicates.

The remainder of this section discusses this result and also addresses the second research question in (8b). Specifically, Section 4.1 discusses a methodological implication of the observation that most of the diagnostics did not yield consistent results when applied to the same contents. Section **LH: fix reference ??** develops a hypothesis for why the two diagnostics in which the target contents were embedded in polar questions yielded greater by-content differentiation than the diagnostics in which the contents were embedded in declaratives. Finally, Section 4.2 addresses research question (8b).

4.1 Methodological implication

The results of Exps. 1–4 differed in the extent to which they replicated previously reported differences among the contents of NRRCs and the complements of five clause-embedding predicates. And even the results of Exps. 5–6, where the target contents were embedded in polar questions, suggested that differences in the diagnostic and the response task can result in differences in the relative at-issueness of target contents.

A methodological implication of the observed diagnostic differences is that diagnostics cannot be treated as interchangeable. Thus, theoretical claims based on one diagnostic should be understood relative to that diagnostic and results obtained with one diagnostic may not generalize to others, as already suggested in Snider 2017a; b; 2018, Koev 2018 and Korotkova 2020. The results of the experiments presented in this paper suggest that empirical investigations of at-issueness must consider that the outcomes of at-issueness diagnostics may depend on the diagnostic used, whether the target content was presented in a declarative sentence or polar question, as well as the response task. As the results of Exps. 5 and 6 suggested that some diagnostics may yield consistent results, a pressing question for future research is whether this also obtains for other pairs of diagnostics and which features of those diagnostics the consistent results can be attributed to.

4.2 Diagnostic differences and notions of at-issueness

Because the diagnostics tested in this paper are grounded in different assumptions about what it means for content to be at-issue, the observed differences between them bear on the broader question of whether diagnostics track a single underlying notion of at-issueness or operationalize distinct phenomena. This subsection discusses how the observed differences relate to patterns that would be expected based on three positions: the unified view, on which different diagnostics target a single unified phenomenon of at-issueness (e.g., Tonhauser 2012; AnderBois et al. 2015; Faller 2019), the dual-notion view, on which they target different underlying notions (Koev 2018); and the anaphoric-accessibility view, on which some diagnostics target at-issueness, while others diagnose the availability of the target content for propositional anaphora (Snider 2017a; a; 2018; Korotkova 2020).**JT: Like I wrote above, I don't understand how the dual-notion and the anaphoric-accessibility views differ since both assume that the various diagnostics target different underlying notions.** These positions differ with respect to how different diagnostics relate to the two definitions of QUD at-issueness and proposal at-issueness, repeated here:

(1) QUD at-issueness: (based on Simons et al. 2010, p. 323)
A content *m* is at-issue in a context *c* iff
a. *m* is relevant[18] to the QUD in *c*, and
b. the speaker has a recognizable intention to address the QUD via *m*.

(2) Proposal at-issueness: (Koev 2013, pp. 50–51)
A proposition *p* is at-issue in a context *c* iff
a. *p* is a proposal in *c* and
b. *p* has not been accepted or rejected in *c*.

The literature differs in how the theoretical assumptions behind these two definitions should be interpreted. And the different positions lead to different expectations about what kind of empirical divergence would count as evidence to support them. The unified view would be supported by convergent results across diagnostics, but it is also consistent with divergence, provided that any diagnostic differences can be explained by factors independent of at-issueness. In contrast, the dual-notion view predicts that at least some diagnostic differences reflect a split between the diagnostics assumed to align with the proposal-based definition (i.e., the dissent-based diagnostics from Exps. 3–4) and those more in line with the QUD-based definition (i.e, the QUD diagnostic from Exp. 1; the 'asking whether' diagnostic from Exps. 2 and 5; and the direct-response diagnostic from Exp. 6). Finally, the anaphoric-accessibility view predicts systematic differences between diagnostics that involve propositional anaphora, like the dissent-based diagnostics (Exps. 3–4) and the direct-response diagnostic (Exp. 6), and those that do not, like the QUD diagnostic (Exp. 1) and the 'asking whether' diagnostic (Exps. 2 and 5).

In the following, we discuss whether our results provide support for any of these views.

The clearest diagnostic difference we found—whether target contents are presented in polar questions or declaratives—may appear related to the distinction between QUD-based and proposal-based definitions of at-issueness, since both involve a contrast between questions and assertions. However, as discussed in § 2.3.3, the QUD-diagnostic (Exp. 1) assumes the QUD-based definition but presents the target content in a declarative sentence, distinguishing between the factors of question embedding and QUD-at-issueness. Contrary to what would be expected based on the dual-notion view, it patterned with the dissent-based diagnostics (Exps. 3 and 4) in showing low by-content differentiation. The observed pattern is also not expected based on the anaphoric-availability view: The direct-response diagnostic from Exp. 6 uses the anaphoric polarity particle *yes* to diagnose at-issueness, but it shows high by-content differentiation and does not pattern with the dissent-based diagnostics which also involve propositional anaphora.**JT: But the anaphoric-accessibility view could be correct and that would still be compatible with the pattern we found because there are so many other factors that could modulate whether the diagnostic produces differentiation. Given all these factors, I'm not sure whether two diagnostics showing the same pattern can be used to argue for anything, or two diagnostics showing a different pattern.** Therefore, the main contrast we found cuts across the groupings expected under views that assume distinct underlying phenomena. As argued in § **??**, this contrast can be explained independently of differences in at-issueness. While all three positions are in principle compatible with some diagnostic differences that are independent of at-issueness, the dual-notion or anaphoric-availability views do lead to expectations about patterns of differences between diagnostics that are not supported here. However, the contrast could be interpreted as most consistent with a unified view, which more directly predicts that any diagnostic differences are due to independent factors.

While not all diagnostic differences we found are explained by the main contrast between question and declarative embeddings, the remaining differences also do not align with the patterns that would be expected based on the different theoretical views. Instead, they appear to reflect diagnostic-specific effects tied to particular contents and response tasks. For instance, the exceptional behavior of *be right* under the QUD diagnostic does not plausibly reflect a difference in at-issueness either. The low ratings more plausibly arise from a pragmatic incompatibility. Our hypothesis here was that *be right* requires a discourse structure that conflicts with the assumptions made by the diagnostic (see discussion in § 2.3.2). As a result, the QUD-diagnostic (Exp. 1) differs from all other diagnostics in its ranking of *be right*. Again, this result is in principle compatible with all three views, but it not lend support to the dual-notion or anaphoric-availability views. Instead, it highlights how diagnostic differences can arise from interactions with expression-specific contextual requirements, which, again, would be the only interpretation that is consistent with the unified view.

Further, some small differences between diagnostics can be attributed to response-task effects. This is evident in the subtle differences between Exps. 5 and 6 (see § 3.2), which differed only in response format, and in our failure to replicate Syrett & Koev's 2015 reported positional effect for appositive NRRCs (see § 2.3.1). While they found sentence-final appositives to be more at-issue than sentence-medial ones under a direct-dissent diagnostic, our dissent-based *yes, but* diagnostic in Exp. 4 showed a difference in the opposite direction, and Exps. 1–3 found no positional difference. Taken together, these results provide no evidence for a stable notion of at-issueness under which sentence-final appositives are systematically more at-issue than medial ones (or the other way round), and they highlight how even small task differences can affect outcomes.

In sum, none of the diagnostic differences we found do not pattern along the lines expected based on the views suggesting that different at-issueness diagnostics are sensitive to distinct underlying phenomena. While this does not provide conclusive evidence to rule out the possibility that some diagnostics are sensitive to distinct underlying notions, it does suggest that much of the observed variation arises from factors other than distinct notions of at-issueness, namely differences in illocutionary force, contextual felicity requirements, and response-task effects.

The anaphoric-accessibility view (Snider 2017b; a; 2018; Korotkova 2020) shares the assumption that differences between diagnostics arise from interactions with expression-specific contextual requirements rather than from differences in at-issueness itself. In particular, it highlights that the direct-dissent and 'yes but' diagnostics rely on propositional anaphora and therefore impose specific constraints on discourse structure, common ground, and anaphoric accessibility. These constraints can independently affect acceptability judgments. However, the direct-response diagnostic (Exp. 6) also involves propositional anaphora (i.e., the polarity particle *yes*) but does not pattern with the dissent-based diagnostics (Exps. 3–4), suggesting that differences between diagnostics cannot always be reduced to whether or not they involve propositional anaphora.

Consequently, our results do not decisively rule out any of the three views discussed here, they can be taken to tentatively suggest that the observed diagnostic differences are most in line with a unified view of at-issueness, where different diagnostics offer different, partially overlapping windows onto a single underlying phenomenon. A final piece of evidence that also tentatively points in favor of this view is that, the diagnostics also showed convergence in the overall rank ordering of contents. For example, *pretend* and *see* consistently ranked low across diagnostics, while *discover* and *know* consistently ranked high. For instance, among most pairs from Exps. 1–4, the Spearman rank correlation was at least moderate (when excluding *be right*, see footnote 13), suggesting some convergence among diagnostics, and that the diagnostics, as implemented there, are at least partially sensitive to a shared underlying property.

5 Conclusion

This paper presented a systematic experimental comparison of at-issueness diagnostics, motivated by two overarching questions: whether different diagnostics yield consistent results when applied to the same contents, and whether any differences between them support the view that they are sensitive to distinct underlying phenomena.

Across six experiments, we tested five diagnostics: the QUD diagnostic, the ‘asking whether’ diagnostic, the direct-dissent diagnostic, the ‘yes, but’ diagnostic, and the direct-response diagnostic. Our results provide experimental confirmation for claims from the theoretical literature that at-issueness diagnostics differ empirically (Snider 2017b; a; 2018; Koev 2018; Faller 2019; Korotkova 2020). Since they did not yield consistent results, they cannot be treated as interchangeable.

We also identified a clear source of variation: diagnostics that present target contents in polar questions showed greater by-content differentiation than those presenting the same contents in declaratives. We argued that this pattern reflects the role of illocutionary force and speaker commitments, and that these factors should be considered when interpreting diagnostic outcomes.

Our findings do not support a view that different diagnostics are sensitive to distinct underlying phenomena. Although the diagnostics diverge empirically, the observed differences did not align with existing distinguishing QUD-based at-issueness, proposal-based at-issueness, or anaphoric availability. Instead, they can be attributed to factors independent of the proposed theoretical distinctions, including speaker commitments, illocutionary force, contextual felicity constraints, and the given alternatives in forced-choice tasks. The results are therefore more consistent with a unified view, on which diagnostics provide partially overlapping perspectives on a single discourse phenomenon that interacts with multiple factors in complex and often subtle ways.

However, our results do not conclusively rule out the possibility that some diagnostics may be sensitive to distinct underlying notions of at-issueness. Determining what different diagnostics can reveal about whether they are sensitive to a shared underlying notion therefore remains an important question for future work.

Data accessibility statement

The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

Ethics and consent

All experiments were conducted with approval from the ethics review committee of [university name redacted for review].

References

Abrusán, Márta. 2011. Predicting the presuppositions of soft triggers. *Linguistics and Philosophy* 34(6). 491–535. https://doi.org/10.1007/s10988-012-9108-y

Abusch, Dorit. 2002. Lexical alternatives as a source of pragmatic presupposition. *Semantics and Linguistic Theory* 12. 1–19. https://doi.org/10.3765/salt.v12i0.2867. Publisher: Ithaca, NY: CLC Publications

Abusch, Dorit. 2010. Presupposition triggering from alternatives. *Journal of Semantics* 27(1). 37–80.

Amaral, Patricia & Roberts, Craige & Smith, E Allyn. 2007. Review of the logic of conventional implicatures by Chris Potts. *Linguistics and Philosophy* 30. 707–749. Publisher: Springer.

Anand, Pranav & Hacquard, Valentine. 2014. Factivity, belief and discourse. In Crni, Luka & Sauerland, Uli (eds.), *The Art and Craft of Semantics: A Festschrift for Irene Heim*, 69–90. MIT Working Papers in Linguistics. https://semanticsarchive.net/Archive/jZiNmM4N/.

AnderBois, Scott & Brasoveanu, Adrian & Henderson, Robert. 2010. Crossing the appositive/at-issue meaning boundary. In *Semantics and linguistic theory*, vol. 20. 328–346. http://journals.linguisticsociety.org/proceedings/index.php/SALT/article/view/2551.

AnderBois, Scott & Brasoveanu, Adrian & Henderson, Robert. 2015. At-issue proposals and appositive impositions in discourse. *Journal of Semantics* 32(1). 93–138. https://doi.org/10.1093/jos/fft014

Atkinson, Maxwell J. & Drew, Paul. 1979. *Order in court*. Houndmills: Macmillan. https://www.researchgate.net/profile/Paul-Drew/publication/230875821_Order_in_Court/links/5e21cd62a6fdcc1015715c1d/Order-in-Court.pdf.

Barnes, Kathryn & Ebert, Cornelia. 2023. The information status of iconic enrichments: modelling gradient at-issueness. *Theoretical Linguistics* 49(3-4). 167–223. https://doi.org/10.1515/tl-2023-2009. https://www.degruyter.com/document/doi/10.1515/tl-2023-2009/html

Beaver, David & Clark, Brady. 2008. *Sense and Sensitivity: How Focus Determines Meaning*. Oxford: Wiley-Blackwell. https://doi.org/10.1002/9781444304176

Brown, Penelope & Levinson, Stephen C. 1978. *Politeness* (Studies in Interactional Sociolinguistics). Cambridge University Press 4th edn.

Bürkner, Paul-Christian. 2017. brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* 80(1). 1–28. https://doi.org/10.18637/jss.v080.i01

Büring, Daniel. 2003. On D-Trees, Beans, And B-Accents. *Linguistics and Philosophy* 26(5). 511–545. https://doi.org/10.1023/A:1025887707652. https://link.springer.com/10.1023/A:1025887707652

Chen, Yuqiu. 2024. *Presuppositions at the Semantics-Pragmatics Interface*: Georg-August-Universität Göttingen Doctoral dissertation. https://ediss.uni-goettingen.de/handle/11858/15164.

Clark, Herbert H. & Schaefer, Edward F. 1989. Contributing to discourse. *Cognitive science* 13(2). 259–294. https://www.sciencedirect.com/science/article/pii/0364021389900086.
Cummins, Chris & Amaral, Patricia & Katsos, Napoleon. 2013. Backgrounding and accomodation of presuppositions: An experimental approach. In *Proceedings of Sinn und Bedeutung*, vol. 17. 201–218.
Degen, Judith & Tonhauser, Judith. 2022. Are there factive predicates? An empirical investigation. *Language* 98(3). 552–591.
Degen, Judith & Tonhauser, Judith. 2025. Projection inferences: On the relation between prior beliefs, at-issueness, and lexical meaning. *Glossa Psycholinguistics* 4(1). https://escholarship.org/uc/item/7hb2h628.
Destruel, Emilie & Onea, Edgar & Velleman, Daniel & Bumford, Dylan & Beaver, David. 2015. A cross-linguistic study of the non-at-issueness of exhaustive inferences. In Schwarz, Floran (ed.), *Experimental approaches to presupposition*, 135–156. Springer. https://doi.org/10.1007/978-3-319-07980-6
Esipova, Maria. 2019. *Composition and projection in speech and gesture*. New York, NY: New York University Doctoral dissertation. https://www.researchgate.net/publication/371702326_Composition_and_projection_in_speech_and_gesture.
Esipova, Maria. 2021. On not-at-issueness in pictures. *Glossa: a journal of general linguistics* 6(1). https://www.glossa-journal.org/articles/10.5334/gjgl.1314/?utm_source=TrendMD&utm_medium=cpc&utm_campaign=Glossa%253A_a_journal_of_general_linguistics_TrendMD_0.
Faller, Martina. 2006. Evidentiality below and above speech acts. Unpublished manuscript. https://semanticsarchive.net/Archive/GZiZjBhO.
Faller, Martina T. 2002. *Semantics and pragmatics of evidentials in Cuzco Quechua (Peru)*: Stanford University Doctoral dissertation. https://www.proquest.com/dissertations-theses/semantics-pragmatics-evidentials-cuzco-quechua/docview/305523331/se-2?accountid=10957.
Faller, Martina T. 2019. The discourse commitments of illocutionary reportatives. *Semantics and Pragmatics* 12. 8:1–53. https://doi.org/10.3765/sp.12.8. https://semprag.org/index.php/sp/article/view/sp.12.8
Farkas, Donka & Bruce, Kim. 2010. On reacting to assertions and polar questions. *Journal of Semantics* 27(1). 81–118. https://doi.org/10.1093/jos/ffp010. https://scholar.archive.org/work/hymiwpqf5fdazop52liak5zuoa/access/wayback/https://people.ucsc.edu/~farkas/papers/assertion_question.pdf
Ginzburg, Jonathan. 1995. Resolving Questions I, II. *Linguistics & Philosophy* 18. 459–527, 567–609.
Ginzburg, Jonathan. 1996. Interrogatives: Questions, facts and dialogue. *The handbook of contemporary semantic theory* 5(18). 359–423. https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=595cd12adcf5e27f900c47778695412e89711481. Publisher: Citeseer.
Ginzburg, Jonathan & Sag, Ivan A. 2000. *Interrogative investigations: The Form, Meaning, and Use of English Interrogatives* (CSLI lecture notes 123). Stanford, CA: CSLI Publications.
Groenendijk, Jeroen Antonius Gerardus & Stokhof, Martin Johan Bastiaan. 1984. *Studies on the Semantics of Questions and the Pragmatics of Answers*: Univ. Amsterdam PhD Thesis.
Gunlogson, Christine. 2001. *True to form: Rising and falling declaratives as questions in English*: University of California Santa Cruz Doctoral dissertation. http://hdl.handle.net/1802/231.
Hofmann, Lisa & de Marneffe, Marie-Catherine & Tonhauser, Judith. 2024. Projection variation: Is the family of sentences really a family? *Sinn und Bedeutung* 28. 422–440.
Horn, Laurence R. 2002. Assertoric inertia and NPI licensing. In *Proceedings of the 38th Annual Meeting of the Chicago Linguistic Society: Parasession on Negation and Polarity*, vol. 38. 55–82.
Hunter, Julie & Asher, Nicholas. 2016. Shapes of Conversation and At-Issue Content. *Semantics and Linguistic Theory* 1022–1042. https://doi.org/10.3765/salt.v26i0.3946. https://journals.

linguisticsociety.org/proceedings/index.php/SALT/article/view/26.1022

Jasinskaja, Katja. 2016. Not at issue any more. *Ms. University of Cologne* https://dslc.phil-fak.uni-koeln.de/sites/dslc/katja_files/jasinskaja_any_more.pdf.

Kakava, Christina. 2002. Opposition in Modern Greek discourse: cultural and contextual constraints. *Journal of Pragmatics* 34(10). 1537–1568. https://doi.org/10.1016/S0378-2166(02)00075-9

Kangasharju, Helena. 2009. Preference for disagreement? A comparison of three disputes. *Talk in interaction: Comparative dimensions* 231–253.

Koev, Todor. 2018. Notions of at-issueness. *Language and Linguistics Compass* 12. e12306. https://doi.org/https://doi.org/10.1111/lnc3.12306

Koev, Todor K. 2013. *Apposition and the structure of discourse*. Rutgers The State University of New Jersey, School of Graduate Studies. https://search.proquest.com/openview/3686668834d9802d690c6e574100c8e0/1?pq-origsite=gscholar&cbl=18750.

Korotkova, Natasha. 2020. Evidential meaning and (not-)at-issueness. *Semantics and Pragmatics* 13. article 4.

Kotthoff, Helga. 1993. Disagreement and concession in disputes: On the context sensitivity of preference structures. *Language in society* 22(2). 193–216. https://www.jstor.org/stable/4168430.

Lee, Jungmee. 2011. *Evidentiality and its interaction with tense: Evidence from Korean*: The Ohio State University PhD Thesis. https://rave.ohiolink.edu/etdc/view?acc_num=osu1306940284.

Lenth, Russell V. 2023. *emmeans: Estimated marginal means, aka least-squares means*. https://CRAN.R-project.org/package=emmeans. R package version 1.8.8.

Locher, Miriam A. 2004. *Power and politeness in action: Disagreements in oral communication*. Walter de Gruyter. https://books.google.com/books?hl=en&lr=&id=wSqi2DbacoUC&oi=fnd&pg=PR5&dq=Locher,+Miriam,+2004.+Power+and+Politeness+in+Action:+Disagreements+in+Oral+Communication.&ots=mCpbw1prD9&sig=JfCGK-J-_5tMnWJRr7N0UOs_13c.

Murray, Sarah E. 2010. *Evidentiality and the structure of speech acts*: Rutgers The State University of New Jersey-New Brunswick Doctoral dissertation.

Murray, Sarah E. 2014. Varieties of update. *Semantics and Pragmatics* 7. 2:1–53. https://doi.org/10.3765/sp.7.2. https://semprag.org/index.php/sp/article/view/sp.7.2

Papafragou, Anna. 2006. Epistemic modality and truth conditions. *Lingua* 116(10). 1688–1702. https://www.sciencedirect.com/science/article/pii/S0024384106000805. Publisher: Elsevier.

Pomerantz, Anita. 1984. Agreeing and disagreeing with assessments: Some features of preferred/dispreferred turn shaped. In Atkinson, J. M. & Heritage, J. (eds.), *Structures of social action*, Cambridge, UK: Cambridge University Press. https://scholarsarchive.library.albany.edu/cas_communication_scholar/3/.

Potts, Christopher. 2005. *The Logic of Conventional Implicatures*. Oxford: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199273829.001.0001

Potts, Christopher. 2015. Presupposition and Implicature. In Lappin, Shalom & Fox, Chris (eds.), *The Handbook of Contemporary Semantic Theory*, 168–202. Wiley 1st edn. https://doi.org/10.1002/9781118882139.ch6. Koev

R Core Team. 2016. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing Vienna, Austria. https://www.R-project.org/.

Roberts, Craige. 2012. Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics* 5. 6:1–69. https://doi.org/10.3765/sp.5.6. https://semprag.org/index.php/sp/article/view/sp.5.6

Sacks, Harvey. 1987. On the preferences for agreement and contiguity in sequences in conversation. In Button, Graham & Lee, John R. E. (eds.), *Talk and social organisation* (Intercommunication series 1), Clevedon, UK: Multilingual matters.

Schlenker, Philippe. 2010. Local contexts and local meanings. *Philosophical Studies* 151. 115–142. https://doi.org/10.1007/s11098-010-9586-0
Scontras, Gregory & Tonhauser, Judith. 2025. Projection without presupposition: A model of clause-embedding predicates. *Sinn und Bedeutung* 29. ???
Scontras, Gregory & Tonhauser, Judith. 2026. Predicting within-utterance projection variation. *Meaning: A journal of Linguistics and Philosophy* 2. 1–27.
Shanon, Benny. 1976. On the two kinds of presuppositions in natural language. *Foundations of language* 14(2). 247–249. https://www.jstor.org/stable/25170057.
Simons, Mandy & Tonhauser, Judith & Beaver, David & Roberts, Craige. 2010. What projects and why. In *Semantics and linguistic theory*, vol. 20. 309–327.
Smithson, Michael & Verkuilen, Jay. 2006. A better lemon squeezer? Maximum-likelihood regression with beta-distributed dependent variables. *Psychological Methods* 11. 54–71.
Snider, Todd. 2017a. At-issueness anaphoric availability. *Proceedings of the Linguistic Society of America* 2. 39–1. http://journals.linguisticsociety.org/proceedings/index.php/PLSA/article/view/4089.
Snider, Todd. 2018. Distinguishing at-issueness from anaphoric potential: A case study of appositives. In *West Coast Conference on Formal Linguistics (WCCFL)*, vol. 35. 374–381.
Snider, Todd N. 2017b. *Anaphoric reference to propositions*. Ithaca, NY: Cornell University Doctoral dissertation. https://ecommons.cornell.edu/server/api/core/bitstreams/3794923d-d85f-4c7c-85ac-8106f83152b5/content.
Solstad, Torgrim & Bott, Oliver. 2024. Cataphoric resolution of projective content: The case of occasion verbs. *Semantics and Pragmatics* 17. 11:1–66. https://doi.org/10.3765/sp.17.11. https://semprag.org/index.php/sp/article/view/sp.17.11
Stalnaker, Robert. 1973. Presuppositions. *Journal of Philosophical Logic* 4. 447–57.
Stalnaker, Robert C. 1978. Assertion. In Cole, Peter (ed.), *Pragmatics* (Syntax and Semantics 9), 315–332. Leiden, Netherlands: Brill. https://doi.org/10.1163/9789004368873_013.
Stalnaker, Robert C. 2002. Common ground. *Linguistics and Philosophy* 25. 701–721. https://www.jstor.org/stable/pdf/25001871.pdf.
Syrett, Kristen & Koev, Todor. 2015. Experimental evidence for the truth conditional contribution and shifting information status of appositives. *Journal of Semantics* 32(3). 525–577. https://doi.org/10.1093/jos/ffu007. Publisher: Oxford University Press
Tannen, Deborah. 1990. You Just Don't Understand: Women and Men. *Conversation. New York: Ballantine books* https://www.researchgate.net/profile/Josep-Maria-Cots/publication/46314186_Tannen_D_1991_You_just_don't_understand_Women_and_men_in_conversation/links/54e70b1b0cf2b19906093ea9/Tannen-D-1991-You-just-dont-understand-Women-and-men-in-conversation.pdf.
Tonhauser, Judith. 2012. Diagnosing (not-) at-issue content. *Proceedings of Semantics of Under-represented Languages of the Americas (SULA)* 6. 239–254.
Tonhauser, Judith & Beaver, David I. & Degen, Judith. 2018. How projective is projective content? Gradience in projectivity and at-issueness. *Journal of Semantics* 35(3). 495–542. https://doi.org/10.1093/jos/ffy007
von Stechow, Arnim. 1991. Focusing and backgrounding operators. In Abraham, Werner (ed.), *Discourse particles: Descriptive and theoretical investigations on the logical, syntactic and pragmatic properties of discourse particles in German*, 37–84. Amsterdam, Netherlands: John Benjamins.
Xue, Jingyang & Onea, Edgar. 2011. Correlation between presupposition projection and at-issueness: An empirical study. In *Proceedings of the ESSLLI 2011 workshop on projective meaning*. 171–184.

Supplements

A Control stimuli in Exps. 1–4

The examples in (1)-(4) provide the two control stimuli used in each of Exps. 1-4. For the a.-examples, participants were expected to give a 'totally fits' response (Exp. 1), a 'yes' response (Exp. 2), a 'totally natural' response (Exp. 3), and a 'no' response (Exp. 4); for the b.-examples, the opposite response was expected. The numbers after each example identify the mean ratings (Exps. 1-3) or the proportion of 'no' responses (Exp. 4) after excluding participants who did not self-identify as native speakers of American English (but before excluding participants on the basis of these controls), showing that the control stimuli worked as intended.

(1) Control stimuli in Exp. 1 (QUD diagnostic)

a. Mary: Which course did Ava take?
John: She took the French course. (.97)

b. Jennifer: What does Betsy have?
Robert: She loves dancing salsa. (.07)

(2) Control stimuli in Exp. 2 ('asking whether' diagnostic)

a. Mary: Did Arthur take a French course?
Question to participants: Is Mary asking whether Arthur took a French course? (.96)

b. Robert: Does Betsy have a cat?
Question to participants: Is Robert asking whether Betsy loves apples? (.02)

(3) Control stimuli in Exp. 3 ('direct dissent' diagnostic)

a. Mary: Arthur took a French course.
Lily: No, he took a Spanish course. (.87)

b. Robert: Betsy has a cat.
Maximilian: No, she doesn't like apples. (.05)

(4) Control stimuli in Exp. 4 ('yes, but' diagnostic)

a. Mary: Arthur took a French course.
Lily: Yes, but Lisa loves cats. / Yes, and he didn't take a French course. / No, he didn't take a French course. (.95)

b. Robert: Betsy has a cat.
Maximilian: Yes, but she is good at math. / Yes, and she loves it so much. / No, she doesn't like apples. (0)

B 20 clauses (items) used in Exps. 5–6

The contents of the following 20 clauses, which instantiated the complements of the 20 clause-embedding predicates, were investigated in Exps. 5–6:

i. Mary is pregnant.
ii. Josie went on vacation to France.
iii. Emma studied on Saturday morning.
iv. Olivia sleeps until noon.
v. Sophia got a tattoo.
vi. Mia drank 2 cocktails last night.
vii. Isabella ate a steak on Sunday.
viii. Emily bought a car yesterday.
ix. Grace visited her sister.
x. Zoe calculated the tip.
xi. Danny ate the last cupcake.
xii. Frank got a cat.
xiii. Jackson ran 10 miles.
xiv. Jayden rented a car.
xv. Tony had a drink last night.
xvi. Josh learned to ride a bike yesterday.

xvii. Owen shoveled snow last winter.
xviii. Julian dances salsa.
xix. Jon walks to work.
xx. Charley speaks Spanish.

C Control stimuli in Exps. 5–6

The control stimuli in Exps. 5–6 were the contents of the main clause polar questions in (5). The non-restrictive relative clauses (NRRCs), given in parentheses in (5), were included in Exp. 6, where at-issueness was measured with an assent diagnostic. The control stimuli here consisted of two clauses (like the target stimuli), to allow the relevant speaker to assent with one of two clauses.

(5) Sentences for control stimuli in in question embedding experiments (Exps. 5–6)
  a. Do these muffins (, which are really delicious,) have blueberries in them?
  b. Does this pizza (, which I just made from scratch,) have mushrooms on it?
  c. Was Jack (, who is my long-time neighbor,) playing outside with the kids?
  d. Does Ann (, who is a local performer,) dance ballet?
  e. Were John's kids (, who are very well-behaved,) in the garage?
  f. Does Samantha (, who is really into fashion,) have a new hat?

We expected participants to give low responses on the at-issueness diagnostics for the control stimuli in (5), indicating that the main clause content is at-issue.